

## Prevalence of xerophthalmia among malnourished children in rural Ethiopia.



To assess the prevalence of eye disease among malnourished children in a rural Ethiopian health center and evaluate correlations between xerophthalmia and grades of malnutrition. A retrospective, cross-sectional survey. An institution-based cross-sectional prospective study was performed at Bushulo Health Center in rural south Ethiopia and included all children age 6 months to 14 years receiving care for malnourishment from June 1st to July 30th, 2008. Data collection involved a combination of interviews with caretakers, ocular examination by a pediatric ophthalmologist and anthropometric measurements. One hundred and seventy-three children (average age at examination  $2.9 \pm 0.2$  years) were treated for malnutrition (97 female, 76 male). One hundred and forty-nine patients had moderate malnutrition (86.03 %) and 24 had severe malnutrition (13.9 %). The following eye diseases were diagnosed--trachoma (12.1 %), blepharitis (13.3 %) and xerophthalmia (20.8 %). Severely malnourished children were more likely to suffer from xerophthalmia than moderately malnourished children ( $p < 0.0001$ ). When comparing anthropometric measurements to the diagnosis of xerophthalmia, only weight percentile showed significance ( $p = 0.008$ ). Xerophthalmia is a common global cause of pediatric blindness and is highly correlated with severe malnutrition. Continued efforts are necessary to improve nutrition and outcomes in these patients.